

TrES-1b

R-0166

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_181022 · created 2026-07-16T17:22:51+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458413.6744 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.014
Transit depth (Rp/Rs)^2	2.2 +/- 0.42 [%]
Orbital Inclination (inc)	89.67 +/- 0.77
Airmass coefficient 1 (a1)	1.024 +/- 0.018
Airmass coefficient 2 (a2)	-0.0137 +/- 0.004
Scatter in the residuals of the lightcurve fit is	1.76 %
Best Comparison Star	#3 - [531, 180]
Optimal Aperture	3.82
Optimal Annulus	8.27
Transit Duration (day)	0.1046 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.148 $\pm$ 0.014 vs 0.1358 (deviation 0.6 σ)
Duration fit vs published	0.1046 d vs 0.1042 d (deviation 0.16 σ)
Mid-time O–C	-4.8 min
Depth SNR	7.3
Residual scatter	1.76 %

Light curve

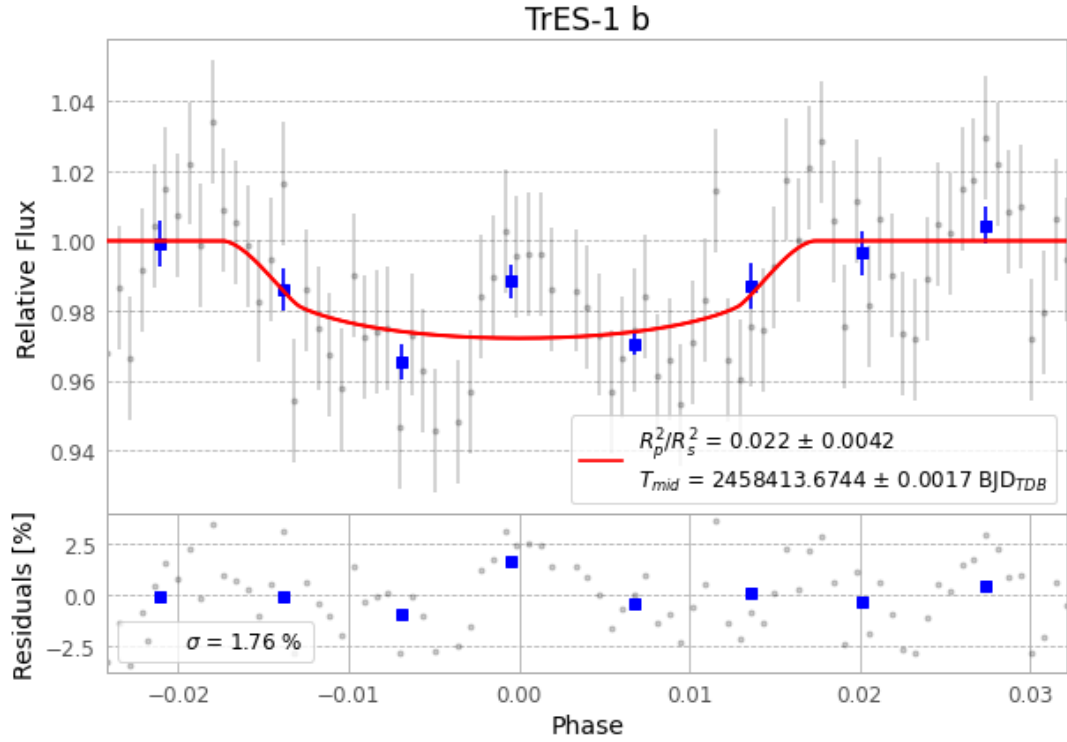


Figure 1 — FinalLightCurve_TrES-1 b_2018-10-21.png (SHA-256 e82f1a0056d4a101...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 341], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 484b71ae13ceb75b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62be8a4

Data provenance

87 FITS frames (57 MB), TRES-1181022021212.FITS → TRES-1181022063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-181022.log (878502287c81...), FinalLightCurve_TrES-1 b_2018-10-21.png (e82f1a0056d4...), FinalLightCurve_TrES-1 b_2018-10-21.csv (955353e4069c...)

Compute resources

Wall time 535.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 17:22Z.